

Mathematical Modeling of Three-Species Ecosystem

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Introduction

The prey-predator model describes the relationship between two groups of animals in nature one prey and another predator. For examples, Deer (Prey) and Wolf (Predator). Predator have higher energy in context of trophic levels. They eat more prey causing the prey population to drop. With less food available, some predators can't survive or have fewer offsprings, so their numbers shrink. This allows the prey to increase their population once again. This prey-predator model shows cyclic behaviour in the ecosystem.

Lotka-Volterra model

It is a mathematical model used to describe how the population of two species one prey another predator interacts. It was developed by Alfred Lotka and Vito Volterra in 1920. These are first order differential equation that model the cyclical dynamics often observed in natural predator-prey systems.

The equation for **prey** is expressed as :

$$\frac{dx}{dt} = \alpha x - \beta xy \quad (1)$$

Where,

x : Population of prey

y : Population of predator

α : Natural growth rate of the prey

β : Rate at which predators kill prey.

The equation for **predator** is expressed as:

$$\frac{dy}{dt} = \delta xy - \gamma y \quad (2)$$

Where,

δ : Rate at which predators increase by consuming prey

γ : Natural death rate of the predators in the absence of food.

Graphical Representation and Analysis

The simulated dynamics of wolves and deer are classic predator–prey cycles. Deer increase first as a function of vegetation availability, then experience a time-lagged increase in wolves as a consequence of ample prey. Enhanced predation brings deer numbers down, followed by a subsequent decline in wolves. This establishes a repeating cycle with a distinct time lag between species peaks.

Graphical Representation and Analysis

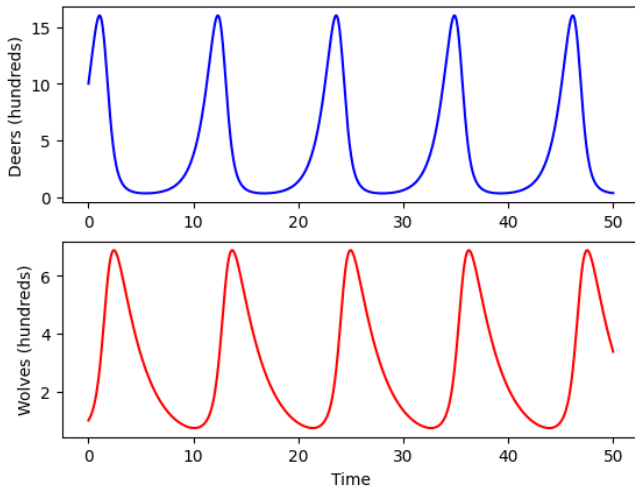


Figure: Prey-Predator interaction

Introduction of 3rd species in the interaction

$G(t)$ = Grass population (or biomass)

$D(t)$ = Deer population

$W(t)$ = Wolf population

The governing principle for each population is

$$\frac{dN}{dt} = \text{Growth} - \text{Loss}, \quad (3)$$

Where, N represents the population of a given species.

Introduction of 3rd species in the interaction

In the absence of herbivores, the grass population grows according to exponential growth,

$$\frac{dG}{dt} = rG, \quad (4)$$

where, r denotes the **intrinsic growth rate**.

But from my standard ecology concepts, We know that environmental resources are limited. So, **Logistic growth** is assumed.

$$\frac{dG}{dt} = rG \left(1 - \frac{G}{K} \right) \quad (5)$$

Where, K represents the carrying capacity

Grass Population

Grass is consumed by deer. Assuming the encounter rate between grass and deer follows the law of mass action,

$$\text{Grass Consumption} = aGD, \quad (6)$$

where, a is the grazing coefficient.

$$\frac{dG}{dt} = rG \left(1 - \frac{G}{K}\right) - aGD \quad (7)$$

$$\boxed{\frac{dG}{dt} = \phi_G^{\text{Growth rate}} - \phi_G^{\text{death rate}}} \quad (8)$$

Deer Population

The deer population changes through three biological processes: reproduction, predation by wolves, natural mortality (natural death rate).

The General equation for the deer population is expressed as:

$$\frac{dD}{dt} = \phi_{\text{birth rate}}^D - \phi_{\text{predation}}^D - \phi_{\text{Natural death}}^D \quad (9)$$

- Deer reproduce by feeding on grass.
- Wolves decrease the deer population through predation.
- Natural mortality also reduces the population.

Wolves Population

The population's growth depends entirely on the availability of deer. When deer are abundant, wolves have enough food to survive and reproduce. No predator is also available. So, the rate of change in the wolf population is the difference between births due to predation and natural deaths.

$$\boxed{\frac{dW}{dt} = \phi_{\text{birth rate}}^W - \phi_{\text{Natural death}}^W} \quad (10)$$

- Birth (Growth): Wolves reproduce by consuming deer. More available deer provide more food, increasing the wolf population.
- Natural Death: Wolves die due to ageing, disease, starvation, or other natural causes.

Graphical representation

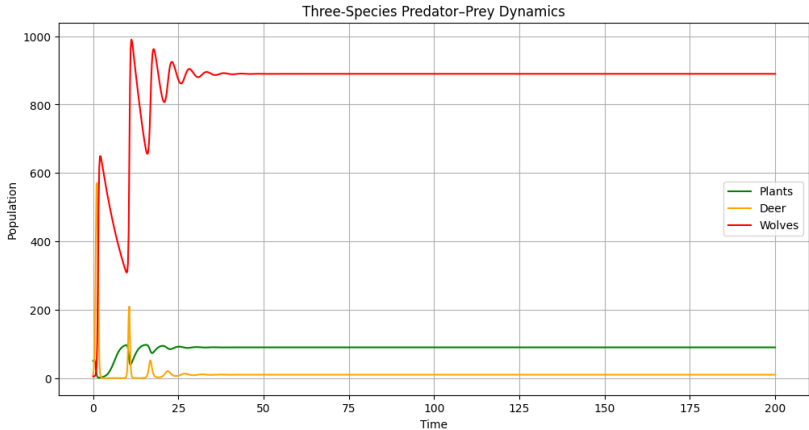


Figure: Three species interaction in the ecosystem

- Initially, the plant population increases rapidly due to low herbivore pressure. This is followed by multiple oscillations in the herbivore (deer) and predator (wolf) populations, with predator peaks lagging slightly behind prey peaks—consistent with classical predator–prey dynamics.
- The final population levels suggest a top-down controlled system: wolves maintain a high population, deer are suppressed to near-extinction levels, and plant biomass stabilizes due to low grazing pressure.

References

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